PV101 Cohort 6 Day 3 - Student Packet

Reading review + wire stripping/crimping lab | 2026-07-08

Name: _____ Group: _____ Date: _____

Today in One Sentence

Today we connect solar basics to real wiring skills: clean stripping, clean crimping, and clean inspection.

Learning Targets

- Explain the difference between sunlight as a resource and electricity as system behavior.
- Identify PV system parts from the Day 1 and Day 2 system path.
- Use power language correctly: volts, amps, watts, and energy over time.
- Prepare de-energized practice wire without damaging strands.
- Make a basic crimp connection and inspect it using a tug test and visual checklist.
- Work in a group of three using rotating roles: Tool Lead, Quality Inspector, Recorder.

Morning Review Notes

Word	Student-friendly meaning	Example or note
PV	Sunlight being converted into electricity.	PV stands for photovoltaic.
DC	Electricity flowing one direction.	Solar modules and batteries are DC.
AC	Electricity that changes direction.	Most home outlets are AC.
Irradiance	Sunlight power right now.	Measured as power per area.
Insolation	Sun energy over time.	Useful for estimating daily production.
Module	A solar panel made of cells.	One panel is one module.
Array	Multiple modules connected together.	Several panels working as a group.
Charge controller	Controls charging from PV to battery.	Protects the battery system.
Inverter	Changes DC to AC.	Used when AC loads are needed.

System Path Practice

Fill in the missing parts:

Sun -> _____ -> _____ -> Battery -> DC Loads

Battery -> _____ -> AC Loads

Power Formula

Watts = _____ x _____

Practice: 12.8 V x 2 A = _____ W

Afternoon Lab Roles

Role	Student name	Job
Tool Lead	_____	Uses stripper/crimper after group agrees.
Quality Inspector	_____	Checks strip length, strands, crimp, and tug test.
Recorder	_____	Writes what passed, failed, and why.

Lab Rule

We are using de-energized practice wire only. Do not connect to batteries, panels, outlets, vehicles, or unknown equipment.

Lab Steps

1. Pick one piece of de-energized practice wire from the instructor-approved bin.
2. Identify the wire size if it is labeled. If not labeled, ask Rich before choosing the stripper slot.
3. Choose a connector that matches the wire size and lab objective.
4. Set the wire in the correct stripper hole. Close the tool gently, pull straight, and remove the insulation.
5. Inspect the exposed copper. Look for cut strands, crushed strands, too much exposed copper, or too little exposed copper.
6. Insert the wire fully into the connector barrel until it seats properly.
7. Place the connector in the correct crimper die and crimp once with firm, controlled pressure.
8. Perform a tug test by pulling the wire and connector in opposite directions with reasonable hand force.
9. Record whether the connection passed or failed and why.
10. Cut off failed practice connections and redo them as practice, not as punishment. The copper does not have feelings.

Connection Inspection Checklist

Check	Pass?	Notes
Correct strip length	[ ] Yes [ ] No	
No missing strands	[ ] Yes [ ] No	
Connector seated	[ ] Yes [ ] No	
Crimp location	[ ] Yes [ ] No	
Tug test	[ ] Yes [ ] No	
Clean finish	[ ] Yes [ ] No	

Practice Record

Attempt	Connector type	Passed tug test?	What changed for next attempt?
1	_____	Yes / No	_____
2	_____	Yes / No	_____
3	_____	Yes / No	_____
4	_____	Yes / No	_____

Exit Ticket

One thing I can do more confidently now:

_____

One thing I still need to practice:

_____

Why does clean wiring matter in an RV solar system?

_____